

(ii) Determine, in terms of  $T$ , the time at which the two samples will have equal activities.

time = .....  $T$  [3]

(c) A radiation detector is placed near to one of the samples in (b).

Explain why the count rate measured by the detector is less than the activity of the sample.

.....  
.....  
..... [2]

[Total: 9]

